



Rwanda's Mara X

Mara intends to turn Rwanda into the next big tech hub, and has released the first made in Africa smartphone <https://digit.in/jan20-45>



Apple's rumoured VR glasses

Patent filings have revealed plans to have hidden optical elements as well as detachable speakers <https://digit.in/jan20-46>

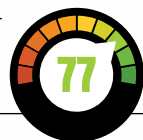
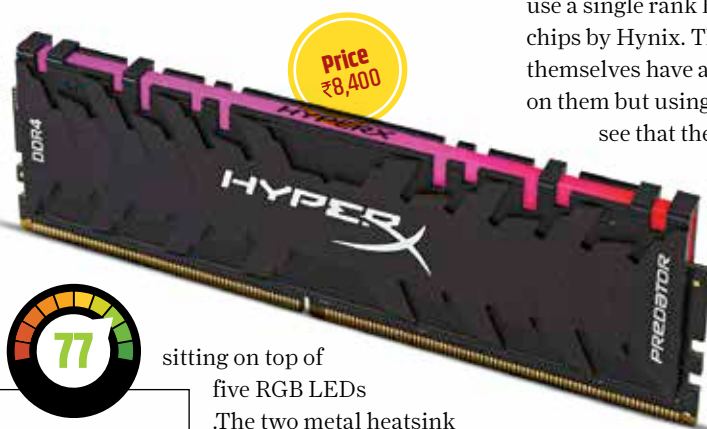
HyperX Predator DDR4 RGB RAM 2933 MHz

Voodoo light synchronisation

The HyperX Predator RGB ram modules are quite different from the run-of-the-mill in the sense that the RGB mechanism on it makes use of infrared light to communicate between individual modules. This way you don't have to install any separate software just to synchronise the modules. However, there's the fact that you won't be able to sync the RGB with the rest of your PC's RGB systems if you let the Infrared Sync do the job alone. Thankfully, it does work with all of the existing RGB lighting systems be it GIGABYTE's RGB Fusion, MSI Mystic Light Sync, ASRock Polychrome RGB or ASUS Aura Sync. If you're wondering whether this added mechanism of IR would be a burden on the power efficiency then you should know that the IR sensor's power characteristics are similar to the individual LEDs that you find on most RGB RAM modules. Except these are passive and don't consume as much power. The technology is patent pending so it'll be awhile before you see it on other manufacturers. We first saw it at COMPUTEX 2019 and there hasn't been another manufacturer with a similar infrared synchronisation technology yet.

Build Quality

The construction of the HyperX Predator RGB modules are not much different from what we've seen before. It makes use of two metal plates for the heatsink which are attached to the PCB and the DRAM chips using thermal tape. The modules are single-sided as is the case with most modules these days so the DRAM chips are all on one side. There's a plastic diffuser



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sitting on top of five RGB LEDs. The two metal heatsink plates are also screwed together. Overall, the modules feel very rugged and well-built.

Coming to the infrared light synchronisation method. The modules have an IR sensor on one side of the module which seeks to communicate with the one that's closest to it and attempt synchronisation. Of all the modules plugged into your system, the one that is farthest from the CPU becomes the master and the remaining modules synchronise with the master module. Should the infrared sensor be blocked in any manner, then the modules switch to static lighting. The reason this manner of synchronisation is superior is that it ignores the memory topology used by the motherboard. Since there is a variation between different board vendors in terms of the memory topology, the sequence in which the memory modules are addressed is different. For example, if there are four modules (1-2-3-4) then depending upon the topology used, the modules might be accessed in sequence or in this manner 1>3>2>4. Now imagine a wave RGB light pattern flowing across the modules. If the topology is not properly read, then you'd have a haphazard display of RGB. This is only helpful if you rely on the infrared sync

technology. If you switch to what the motherboard vendor is using, you're on your own.

Performance

The HyperX Predator RGB modules use a single rank hierarchy with the chips by Hynix. The DRAM chips themselves have a Kingston labelling on them but using ThetaPhoon we can see that these are Hynix chips and they run at 1.2v using CL15-17-17-39 timings. In AIDA64's cache and latency test, the modules clock in about 73.4ms at 2933 MHz and as we overclock the

DIMMs, we can see it drop to 69 @ 3200 MHz, 67 @ 3333 MHz and 66 @ 3400 MHz. We even managed to overclock the DIMMs to 3600 MHz but couldn't complete all the benchmarks at that clock speed. In the Memory read benchmark, we managed to get about 43,134 MB/s and during the write test it scored 45,655 MB/s. At 3200 MHz, the read throughput rose to 49,276 MB/s and write speeds were at 49,212 MB/s.

Verdict

Given their unique RGB light synchronisation technology, these modules really stand apart from the rest of the manufacturers. The units are also very well built and the Hynix DRAM chips used are quite formidable as well given that our kit managed to scale to 3400 MHz with ease and even hit 3600 MHz. The heatsink isn't gimmicky and pricing is very competitive. Overall, the HyperX Predator RGB RAM modules are worth the money.

—Mithun Mohandas

SPECIFICATIONS

TYPE: DDR4 | CAPACITY: 2x 8 GB | RATED SPEED: 2933 MHz | TIMINGS: 15-17-17-39 | OEM: Hynix | PART NO: HX429C15PB3A/8 | RANKS: 1 | BANKS: 16.

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